CBRC Carl Balita Review Center

OUR DAILY BREAD

Romans 8: 31
"If God is for us, who can be against us?"

PHYSICAL SCIENCE

METRIC SYSTEM

- Makes use of the base ten place value system
- Convert from one metric measure to another by multiplying or dividing be ten or moving a decimal place.

Prefixes used in the Metric System:

- **Meter –** basic unit for length (Km)
- **Grams –** basic unit for mass (cg)
- **Liters –** basic unit for volume (mL)

Prefix	Symbol	Numeral Form	Scientific notation
nano	n	0.000000001	1×10^{-9}
micro	μ	0.000001	1×10^{-6}
milli	m	0.001	1×10^{-3}
centi	c	0.01	1×10^{-2}
deci	d	0.1	1×10^{-1}
deka	da	10	1 x 10
hecto	h	100	1×10^{2}
kilo	k	1000	1×10^{3}

The SI Units of Measurement

English-Metric		Metric-English		Length	
1 in	2.54 cm	1 cm	0.3937 in	12 in	1 ft
1 ft	0.3048 m	1 m	3.281 ft	3 ft	1 yd
1 yd	0.9144 m	1 m	10.94 yd	5280 ft	1 mile
1 mile	1.609 km	1 km	0.6214 mi	**Weight**	
1 qt	0.946 L	1 L	1.057 qts	16 ou	1 lb
1 gal	3.785 L	1 L	0.2642 gal	200 lbs	1 ton
1 ou	28.35 g	1 g	0.0353 ou	**Volume**	
1 lb	453.59 g	1 g	0.0022 lb	2 cups	1 pt
				4 qt	1 gal

SCALAR AND VECTOR QUANTITIES

1. **SCALAR:** Quantities signifying magnitude only
   - **Ex:** Mass, charge, length, temperature, speed

2. **VECTOR:** Quantities signifying magnitude and direction
   - **Ex:** Weight, displacement, Velocity, Acceleration, Momentum
   -

- **Vector Addition:** Computing for the resulting or net magnitude of vector quantities
  - **Upward** and **Right** motion indicates positive (+) sign
  - **Downward** and **Left** motion indicates negative (-) sign

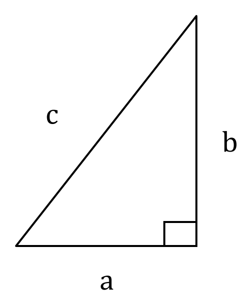
5 (→) + 5 (→) = 10 (→)
5 (→) + 5 (←) = 0
5 (→) + 10 (--→) = 15 (----→)
5 (→) + -10 (←--) = -5 (←)
5 (→) + -15 (←----) = -10 (←--)

Example: A care moves 20 km north, then 10 km south. What is the displacement of the car?

- **Solution:** 20 km + 10 km northward (since the resultant has positive sign it indicates a direction towards north)

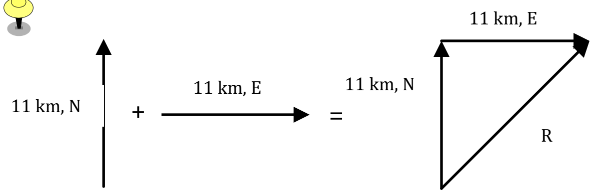
PYTHAGOREAN THEOREM: to determine the result of adding ONLY TWO vectors that make a RIGHT ANGLE to each other.

$a^2 + b^2 = c^2$



Example: James leaves the base camp and hikes 11 km, north and then hikes 11 km east. Determine James' resulting displacement.

- **Solution**: The result (resultant) of walking 11 km north and 11 km east is a vector directed northeast as shown in the diagram to the right. Since the northward displacement and the eastward displacement are right angles to each other, the Pythgorean theorem can be used to determine the resultant.



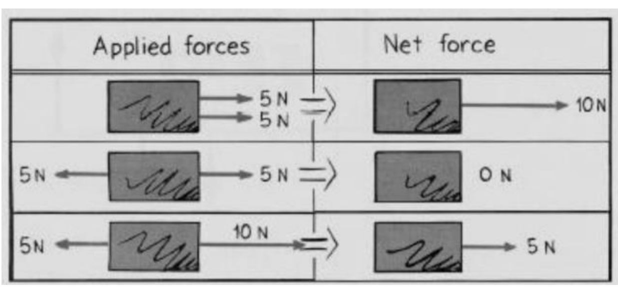
$$11^2 + 11^2 = R^2$$
$$242 = R^2$$
$$156 = R$$

The result of adding 11 km north plus 11 km east is a vector with a magnitude of 15.6 km.

NEWTON'S LAWS OF MOTION

1. **FIRST LAW:** Every object continues in its state of rest, or of uniform motion with constant speed in a straight line, unless acted upon by unbalanced external forces impressed upon

Inertia	Net Force	Normal Force
The property of things to resist changes in motion	The vector sum of forces that act on an object	The force equal in magnitude but opposite in direction of the gravitational force



- **Mechanical Equilibrium –** state of an object which there are no changes in motion
  - If at rest, the state of rest persists
  - If moving, motion continues without change

  Objects at equilibrium have net force of ZERO. (In the diagram above, the second example is in mechanical equilibrium)

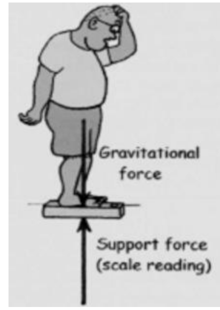
Example:

1. What is the net force on a bathroom scale when a 50-kg person stands on it?
2. Suppose you stand on two bathroom scales with your weight evenly divided between the two scales. What will each scale read? What happens when you stand with more of your weight on one foot than the other?

Answer:

- Gravitational force is equal to the support force!

1. Zero, as evidenced by the scale remaining at rest. The scale reads the support force, which has the same magnitude as weight- not the net force.



2. The reading on each scale is half your weight. Then the sum of the scale readings will balance your weight and the net force on you will be zero. If you lean more on one scale than the other, more than half your weight will be read on that scale but less on the other, so they will still add up to your weight.

2. **SECOND LAW OF MOTION:** When the resultant or net force acting on an object is not equal to zero, the object will accelerate.

$$\mathbf{F = ma}$$

where F – force in Newtons
m – mass in Kg
a – acceleration in m/s^2

- **Acceleration** is directly proportional to force (as one increases, the other increases) but inversely proportional to mass (as one increases, the other decreases)

Mass	The quantity of matter in an object
Weight	The force due to gravity on an object
Newtons	The SI unit of force. One newton (N) is the force that will give an object of mass 1 kg an acceleration of 1 m/s^2
Volume	The quantity of space an object occupies

Example:

1. Find the acceleration of a 3.0 kg object when a net force of 30N acts on it?

$$F = ma$$
$$30N = 3.0\text{ kg} \times a$$
$$a = 30N / 3.0\text{ kg}$$
$$a = 10\ m/s^2$$

2. A 3 kg object requires 10N of force to accelerate it at a certain speed. How much force will a 6 kg object require to accelerate it at the same speed?
   - Since the mass is directly proportional to force, as mass increases the force required also increases. The 6 kg object has twice mass from that of the 3 kg object so it would require twice as much force to achieve the same acceleration. Twice 10N is 20N.

3. **THIRD LAW OF MOTION:** To every action there is always an opposed equal reaction.
   - Whenever one object exerts a force on a second object, the second object exerts an equal and opposite force on the first.

Example: While driving down the road, a firefly strikes the windshield of a bus and makes a mess in front of the bus. The firefly hit the bus and the bus hits the firefly. Which of the two forces is greater: the force on the firefly or the force on the bus?

Answer: The forces on the fly and on the bus are EQUAL.

Uniformly Accelerate Motion	
Distance	How far one object moves from location to another
Displacement	Distance with direction
Speed	Distance traveled per unit of time; measures how fast an object changes position
Velocity	Speed of an object with direction
Acceleration	Rate at which velocity changes with time, in magnitude or direction

Speed = $\frac{distance}{time}$

Average speed = $\frac{total\ distance\ covered}{time\ interval}$

Acceleration = $\frac{Change\ of\ velocity}{time\ interval}$

ENERGY, WORK, POWER

2. **Energy –** Property of a system that enable it to do work.
   - **Potential Energy:** Energy at rest; Energy that something possesses because of its position

     weight x height = (m)(g)(h)

   - **Kinetic Energy:** Energy in motion

     mass x speed^2

   - **Mechanical Energy:** Energy due to the position of something or the movement of something
   - **Conservation of Energy:** Energy cannot be created or destroyed, only transformed from one form into another, but total amount of energy never changes.
3. **Work –** Product of force and the distance moved; unit of work is joule

   Work = force (f) x distance (d)
4. **Power –** Rate at which energy is expended; unit of power is joule/sec.

   Power = work done (W)/time (t)

THERMODYNAMICS – study of heat and its transformation to different forms of energy.

- **Internal Energy:** Energy produced from the attractive and repulsive forces of molecules in an object which increased as temperature increases
- **Heat/ Thermal Energy –** Energy produced as heat is transferred from object with higher temperature to that with lower temperature until it reaches equilibrium.

Methods of Heat Transfer:

1. **Conduction:** transmission of heat from two objects with DIRECT contact
2. **Convection:** heat transfer through AIR and water currents
3. **Radiation:** heat transfer through RAYS or WAVES emitted by a very hot object.

Temperature: measure of the average translational kinetic energy per molecule in a substance, measured in degrees Celsius, Fahrenheit or Kelvin

Celsius to Fahrenheit = (°C x $\frac{9}{5}$) + 32

Fahrenheit to Celsius = (°F - 32) x $\frac{5}{9}$

Celsius to Kelvin = °C + 273.15

- **Absolute zero:** lowest possible temperature that a substance may have
- **Newton's Law of Cooling:** rate of loss of heat from an object is proportional to the temperature difference between object and its surroundings